Programming Questions and Answers

Question 1: What is Java?

Answer: Java is a high-level, object-oriented programming language. It is platform independent because Java programs run on the Java Virtual Machine (JVM).

Question 2: What is a Class?

Answer: A class is a blueprint or template used to create objects. It defines the data (fields) and behavior (methods) of objects.

Question 3: What is an Object?

Answer: An object is an instance of a class. Objects have state (data) and behavior (methods).

Question 4: What is OOP?

Answer: OOP (Object-Oriented Programming) is a programming paradigm based on classes and objects. Its four main principles are encapsulation, inheritance, polymorphism, and abstraction.

Question 5: What is JVM?

Answer: The Java Virtual Machine (JVM) is responsible for executing Java bytecode. It allows Java applications to run on different operating systems without recompilation.

Question 6: What is Inheritance?

Answer: Inheritance is a feature in which one class acquires the properties and methods of another class. It promotes code reuse.

Question 7: What is Polymorphism?

Answer: Polymorphism allows the same method or interface to behave differently depending on the object that invokes it.

Question 8: What is Encapsulation?

Answer: Encapsulation is the process of wrapping data and methods together into a single unit and restricting direct access to the data.

Question 9: What is Abstraction?

Answer: Abstraction hides implementation details and exposes only the essential functionality to the user.

Question 10: Why is Java platform independent?

Answer: Java code is compiled into bytecode, which is executed by the JVM. Since the JVM is available on multiple operating systems, the same Java program can run on Windows, Linux, and macOS.